# Shalini Rai


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## EDUCATION

[ OCT 2022 – JAN 2026 ]

**Master of Science** / Electrical Engineering
***University of Stuttgart, Stuttgart***

- Early Detection of Thermal Runaway with Signal Processing: Developed correlation-based signal processing methods on voltage, current, and temperature data from a 10-cell Li-ion stack to detect and classify incipient faults (internal short, contact, sensor, and thermal faults). Implemented a sliding-window diagnostic pipeline with a low-cost thermal model for robust early fault detection.
- Investigation of Neural Equalization-Enhanced Phase Noise Mitigation: Investigated CNN-based equalizers for EEPN mitigation in coherent optical links under varying dispersion, linewidth, and distance. Analyzed generalization limits (jail-window effect) and evaluated entropy-regularized loss and AIR-based early stopping to improve robustness and achievable information rate.

[ AUG 2018 – FEB 2019 ]

**Post Graduate Diploma** / Big Data Analytics
***Centre for Development of Advanced Computing, Pune***

- Vehicular Crash Risk Prediction: Performed lane-level risk mapping using geospatial ML models and Hadoop HDFS for large-scale roadway datasets (85.32% accuracy).

[ AUG 2013 – JUL 2017 ]

**Bachelor of Technology** / Electronics and Communication Engineering
***Dr. APJ Abdul Kalam Technical University, Lucknow***

- Real-time Information-based Air Quality Index Monitor: Designed an Arduino-based Air Quality Monitor for pollutant detection and ppm estimation, with processing and GSM transmission programmed via MATLAB.

## WORK EXPERIENCE

[ MAR 2024 – MAR 2026 ]

**Working Student, Software Engineering**
***Philips, Boeblingen***

- Developed a Python-based ML pipeline to classify damping in invasive arterial blood pressure (IBP) signals for hemodynamic analysis.
- Automated data processing workflows incorporating feature engineering and a patient-criteria-based train/test splitting strategy; documented methodology and performance results for R&D review.
- Built PyQt-based GUIs and protocol logging pages for clinical study tools, integrated with system-level testing to ensure software reliability.
- Applied Clean Code and TDD (Test-Driven Development) principles within a CI/CD environment (Git/GitHub) to deliver production-quality software.
- Annotated IBP datasets by filtering corrupt signals and marking clinically relevant episodes for model training and data insights.

[ JUL 2021 – SEP 2022 ]

**Senior Assistant, Data Science**
***Deloitte, Bengaluru***

- Developed an NLP framework for Audit Automation, implementing Named Entity Recognition (NER) to extract entities (designations, locations, financial amounts) and NLG to generate automated narratives for internal control reporting.
- Collaborated with internal Audit SMEs to develop a tool for deployment across internal audit teams.
- Performed journal entry anomaly detection and regularity analysis on financial data to identify potential control and audit issues.

[ MAY 2019 – JUN 2021 ]

**Junior Data Scientist**
***Xcaliber Infotech Pvt Ltd, Pune***

- Implemented sentiment analysis on high-volume conversational text and discussion-based datasets using BERT (PyTorch).
- Built automated web scraping and price forecasting pipelines for consumer electronics across primary e-commerce sites and secondary market platforms.
- Deployed ML models via Flask API on Linux servers to provide real-time predictive analytics for telecom clients.
- Developed a computer vision system for cosmetic image grading using OpenCV and TensorFlow for image preprocessing and visualization.

## PROFESSIONAL ACTIVITIES

[ SEP 2020 & DEC 2021 ]

**Namibia University of Science & Technology**, ***Virtual***
Delivered guest lectures on Distributed File Systems

[ JUL 2020 & DEC 2020 ]

**Volunteer**, ***Virtual***
Conference on Neural Information Processing Systems (NeurIPS 2020)
International Conference on Machine Learning (ICML 2020)

## SKILLS & LANGUAGES

**AI & Machine Learning**: LLMs | Deep Learning (PyTorch, TensorFlow) | NLP (BERT) | Computer Vision (OpenCV) | Statistical Signal Processing
**Engineering & MLOps**: Python | C++ | Docker | CI/CD | TDD | SQL | Linux | Git | Azure
**Electrical Core**: Optical and Wireless Communications | Photovoltaics | Information Theory | Embedded Systems | Sensor Principles
**Languages**: English (Professional) | German (B1.1) | Hindi (Native)